

Universal Stellar Control of Planetary Spacing – Mathematical Framework

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Overview

This document provides a step-by-step, fully explicit mathematical procedure for computing the expected orbital spacing of a planetary system from the effective temperature of its host star. The core result is a universal relation between stellar effective temperature $T_\star$, the logarithmic spacing parameter β , the geometric spacing factor q , and the semi-major axis ladder $\{a_n\}$.

$$T_\star \Rightarrow \beta(T_\star) \Rightarrow q(T_\star) = e^{\beta(T_\star)} \Rightarrow a_n = a_0 q^n$$

The key ingredient is a sigmoid (logistic) function for $\beta(T)$, derived from disk thermodynamics, Hill stability, and migration saturation limits.

Step 1 – Input: Stellar Effective Temperature

1.1. Define the stellar parameter

For any main-sequence star, the only required input to this model is the stellar effective temperature:

$$T_\star \quad (\text{in Kelvin})$$

This value is typically obtained from stellar spectroscopy, photometry, or catalog data.

Step 2 – Compute the Logarithmic Spacing Parameter $\beta(T_\star)$

2.1. Physical meaning of β

The parameter β is defined as the logarithm of the geometric spacing factor q between adjacent planetary orbits:

$$\beta = \ln q$$

A larger β corresponds to wider logarithmic spacing (larger q), while a smaller β corresponds to more tightly packed orbits.

2.2. Universal sigmoid law for $\beta(T)$

The model posits that β is a smooth, bounded function of stellar effective temperature, constrained by a minimum packing floor and a maximum packing ceiling. The functional form is:

$$\beta(T) = \beta_{\min} + \frac{\beta_{\max} - \beta_{\min}}{1 + \exp\left(\frac{T - T_0}{\Delta}\right)}$$

2.3. Parameter values

The parameters are set by physical considerations and calibrated against observed systems:

- **Minimum spacing (hot-star floor):** $\beta_{\min} \approx 0.09$
- **Maximum spacing (cool-star ceiling):** $\beta_{\max} \approx 0.30$
- **Transition midpoint temperature:** $T_0 \approx 4800$ K
- **Transition width:** $\Delta \approx 700$ K

Substituting these into the general expression yields the explicit working formula:

$$\beta(T) = 0.09 + \frac{0.22}{1 + \exp\left(\frac{T - 4800}{700}\right)}$$

2.4. Step-by-step evaluation for a given star

1. **Compute the normalized temperature offset:**

$$x = \frac{T_{\star} - 4800}{700}$$

2. **Compute the exponential term:**

$$E = \exp(x) = \exp\left(\frac{T_{\star} - 4800}{700}\right)$$

3. **Compute the logistic denominator:**

$$D = 1 + E$$

4. **Compute the logistic fraction:**

$$F = \frac{0.22}{D}$$

5. **Compute the final β :**

$$\beta(T_{\star}) = 0.09 + F$$

Step 3 – Compute the Geometric Spacing Factor $q(T_{\star})$

3.1. Definition

Once $\beta(T_{\star})$ is known, the geometric spacing factor q is obtained via:

$$q(T_{\star}) = \exp(\beta(T_{\star}))$$

3.2. Step-by-step evaluation

1. **Take the exponential of β :**

$$q = e^{\beta(T_{\star})}$$

2. **Interpretation:** this is the factor by which each orbit is typically larger than the previous one in a logarithmically spaced planetary system.

Step 4 – Constructing the Orbital Ladder $\{a_n\}$

4.1. Geometric progression of semi-major axes

The semi-major axes of planets in a system that follows this universal spacing law can be written as:

$$a_n = a_0 q^n$$

where:

- a_0 is a chosen reference radius (e.g., the innermost planet or a characteristic inner edge).

- q is the spacing factor computed from the stellar temperature.
- n is an integer index (0, 1, 2, ...) labeling successive planets.

4.2. Step-by-step construction

1. Choose a reference radius a_0 :

This may be:

- The observed semi-major axis of the innermost planet, if known.
- A theoretically motivated inner edge (e.g., dust sublimation radius, corotation radius).

2. Compute successive orbits:

$$a_0, \quad a_1 = a_0 q, \quad a_2 = a_0 q^2, \quad a_3 = a_0 q^3, \quad \dots$$

3. Compare with observed planets:

Match observed semi-major axes to the nearest rung of the ladder to:

- Identify missing planets (gaps in the ladder).
- Quantify deviations due to migration or scattering.

Step 5 – Worked Example (Generic Star)

As an illustration, consider a star with effective temperature:

$$T_{\star} = 5000 \text{ K}$$

5.1. Compute $\beta(T_{\star})$

1. Normalized offset:

$$x = \frac{5000 - 4800}{700} = \frac{200}{700} \approx 0.2857$$

2. Exponential:

$$E = \exp(0.2857) \approx 1.331$$

3. Denominator:

$$D = 1 + 1.331 \approx 2.331$$

4. Logistic fraction:

$$F = \frac{0.22}{2.331} \approx 0.0944$$

5. Final β :

$$\beta(5000) = 0.09 + 0.0944 \approx 0.1844$$

5.2. Compute $q(T_\star)$

$$q = e^{0.1844} \approx 1.202$$

5.3. Build the orbital ladder

Suppose we choose $a_0 = 0.05$ AU as the inner reference radius. Then:

$$a_0 = 0.05 \text{ AU}$$

$$a_1 = 0.05 \times 1.202 \approx 0.0601 \text{ AU}$$

$$a_2 = 0.05 \times 1.202^2 \approx 0.0723 \text{ AU}$$

$$a_3 = 0.05 \times 1.202^3 \approx 0.0869 \text{ AU}$$

$$a_4 = 0.05 \times 1.202^4 \approx 0.1044 \text{ AU}$$

...

These radii define the expected geometric ladder for a planetary system around a 5000 K star, assuming it follows the universal spacing law at the baseline (minimum-spacing) level.

Step 6 – Interpretation and Extensions

6.1. Baseline vs. migrated systems

The $\beta(T)$ curve derived here represents a *minimum-spacing baseline* set by stellar and disk physics. Real systems may exhibit:

- $\beta_{\text{obs}} \approx \beta(T_\star)$: baseline, minimally migrated systems.
- $\beta_{\text{obs}} > \beta(T_\star)$: more compact, strongly migrated or resonance-packed systems.

Crucially, systems are not expected to lie *below* the curve, as that would violate Hill stability and dynamical packing limits.

6.2. Predictive applications

- Predicting typical orbital spacing for any star from its spectrum.
- Identifying missing planets as gaps in the geometric ladder.
- Classifying systems as baseline vs. migration-enhanced architectures.

- Constraining formation and migration histories from deviations in β .